

How to write an R Markdown file

A short guide, using lecture03_tidyverse.Rmd as the example

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1 What an R Markdown file is

An R Markdown file (`.Rmd`) is a text file that mixes three things:

1. a **header** at the top, between two lines of `---`, that says what the document is called and what it should become (PDF or web page);
2. **text**, written in Markdown, a plain way of marking headings, bold and lists;
3. **code chunks**, blocks of R code that run when the document is *knitted*, with their output placed right after them.

Pressing **Knit** in RStudio turns the file into a finished document: your explanation, your code and its results, together. That is why assignments should use it: the reader sees exactly what you ran and what came out.

This guide is itself an R Markdown file. Open `rmarkdown_guide.Rmd` next to the PDF and compare them; every part described below is visible in the source.

2 Create one

In RStudio: **File** → **New File** → **R Markdown...** Give it a title and your name, choose **PDF** or **HTML**, click **OK**. RStudio opens a file with example content. Delete everything below the header and write your own.

Save it with **Ctrl + S**, in a folder of its own, with any data in a `data/` subfolder next to it. When knitting, R uses the folder of the `.Rmd` file as its working directory, so `read_csv("data/students.csv")` finds the file with no `setwd()`.

3 The three parts, using the lecture file

3.1 The header

The top of `lecture03_tidyverse.Rmd` looks like this:

```
---
title: "Lecture 3: Introduction to the Tidyverse"
author: "Touseef Hameed"
date: "16 September 2026"
output: pdf_document
---
```

Change the title, author and date. `output: pdf_document` makes a PDF; `output: html_document` makes a web page. Keep the indentation exactly as shown; the header is sensitive to spaces.

3.2 Text

Ordinary sentences are just typed. A few marks give structure:

```
# A main heading
## A smaller heading
```

Plain paragraph. **Bold**, *italic*, and `code in the text`.

- a bullet point
- another one

1. a numbered step
2. the next step

A link: <https://r4ds.hadley.nz>

In an assignment, use one `#` heading per problem, then the code, then your explanation.

3.3 Code chunks

A chunk starts with ````{r}` on its own line and ends with `````:

```
```{r}
library(tidyverse) # load the packages
students <- read_csv("data/students.csv") # read the data
head(students) # first six rows
```
```

Insert a new chunk with **Ctrl + Alt + I**. Run one chunk without knitting by clicking the green **play** button at its top right. Its output appears under it in the editor.

When the document is knitted, that chunk produces this (the long message that `library(tidyverse)` prints is hidden here with `message=FALSE`, see Section 4):

```
library(tidyverse)
students <- read_csv("data/students.csv", show_col_types = FALSE)
head(students)
```

```
#> # A tibble: 6 x 5
#>   'Student ID' 'Full Name' favourite.food mealPlan      AGE
#>   <dbl> <chr>          <chr>          <chr>    <chr>
#> 1         1 Sunil Huffmann Strawberry yoghurt Lunch only      4
#> 2         2 Barclay Lynn   French fries    Lunch only      5
#> 3         3 Jayendra Lyne  N/A            Breakfast and lunch 7
#> 4         4 Leon Rossini    Anchovies       Lunch only    <NA>
#> 5         5 Chidiegwu Dunkel Pizza           Breakfast and lunch five
#> 6         6 Güvenç Attila    Ice cream       Lunch only      6
```

4 Chunk options you will need

Options go inside the braces after `r`, separated by commas.

| option | effect | use it for |
|----------------------------|---|---|
| <code>eval=FALSE</code> | show the code, do not run it | <code>install.packages("tidyverse")</code> , which must not run at every knit |
| <code>echo=FALSE</code> | run the code, hide it, show only the output | a plot where the code is not the point |
| <code>include=FALSE</code> | run the code, show nothing | a setup chunk |

| option | effect | use it for |
|---------------------------------|----------------------------|--|
| message=FALSE,
warning=FALSE | hide messages and warnings | the long message that
library(tidyverse) prints |

Example, as written in the lecture file:

```
```{r, eval=FALSE}
install.packages("tidyverse") # run once on each computer you use
```
```

5 Knit

Click **Knit** (the ball-of-wool icon above the editor), or press **Ctrl + Shift + K**. The document is built in a **fresh R session**: nothing from your console is available. Every package must be loaded and every object created *inside* the document, in order.

The result is saved next to the .Rmd file, with the same name and .pdf or .html.

PDF needs a LaTeX installation. If knitting to PDF fails, either run once in the console

```
install.packages("tinytex")
tinytex::install_tinytex()
```

or change the header to output: html_document. HTML is accepted for the assignment.

6 The errors everyone meets

| message | cause | fix |
|--------------------------------------|---|--|
| object 'students' not found | the chunk that creates it is missing, or comes later in the file | create every object before you use it, inside the document |
| could not find function "read_csv" | library(tidyverse) is not in the document, or comes after the chunk | put it in the first chunk |
| cannot open file 'data/students.csv' | the file is not in a data folder next to the .Rmd | move it there, or use the web address |
| the knit stops at install.packages | the chunk is being run | add eval=FALSE to that chunk |
| Error in parse: unexpected symbol | a typing error in the chunk, often a missing comma or quote | run the chunk alone with the play button to find the line |

The knitting log, in the **R Markdown** tab at the bottom of RStudio, names the chunk where the problem is.

7 For the assignment

- One heading per problem: `# Problem 1`, `# Problem 2`, and so on.
- Under each: the chunk with the code and, for the written problems, your text.
- A comment on every command.
- Knit it, check that the output under each chunk is what you meant, and upload the `.Rmd` together with the knitted `.pdf` or `.html`.

If knitting does not work on your computer, screenshots of your RStudio window showing the code and its output are accepted instead.